

The Saudi Legal Corpus for AI: An Auditable, Official-Source-Grounded, Article-Level Corpus of Saudi Arabian Legislation

Abdullah Almohammed

Independent Researcher

abdullah.m.almohammed@gmail.com

ORCID: 0009-0001-0832-0995

Abstract

Legal natural language processing has advanced rapidly for English and several European languages, but Arabic — and Saudi Arabian law in particular — remains critically under-resourced: to our knowledge, no openly documented, machine-readable, article-level corpus of Saudi legislation has previously been described. We present the Saudi Legal Corpus for AI, a corpus of 290 legislative instruments (201 statutory laws, 64 statutory regulations, 24 implementing regulations, and one set of procedural rules) covering the Kingdom of Saudi Arabia’s principal legislation, structured into 15,689 article-level records (~ 1.2 million Arabic tokens) in a unified, retrieval-ready index spanning twelve legal domains. The corpus rests on three design commitments that distinguish it from web-scraped legal collections: official-source grounding, in which every track records its issuing authority and official source and every record carries its own provenance label (251 distinct labels) under a four-tier source-verification taxonomy; auditability, enforced by 376 read-only, idempotent validators and fully deterministic derived layers; and the governing-text principle, under which the official Arabic text governs and all other language layers are explicitly non-authoritative. Beyond the text, the corpus ships analytical layers that are, to our knowledge, the first of their kind for Saudi law: a statutory citation graph (3,607 references, 585 of them between distinct instruments), a hand-classified supersession graph (102 edges), a glossary of 1,920 statutorily defined terms with 3,318 definitions, an embedding-ready chunking layer (16,304 chunks), and a manually confirmed retrieval benchmark of 519 gold queries. On that benchmark the corpus’s metadata-based lexical searcher reaches 93.3% top-1 accuracy against 90.4% for a standard BM25 baseline, with the gap concentrated on definitional queries (90.9% versus 74.5%), quantifying the value of the engineered retrieval metadata while leaving informative headroom for neural Arabic legal retrieval. We describe the construction methodology in full, report statistics that are reproducible from the archived release, and discuss intended uses, limitations, and the legal and ethical boundaries of releasing structured national legislation for artificial-intelligence research.

Keywords: Legal NLP · Arabic language resources · Saudi Arabia · Legislative corpus · Information retrieval · Data provenance

1 Introduction

Large language models are increasingly consulted for legal questions, and retrieval-augmented generation (RAG) over authoritative legal text is the standard mitigation for hallucinated law. That mitigation is only available where authoritative legal text exists in machine-readable, retrieval-ready form. For the law of the Kingdom of Saudi Arabia — a G20 economy that has recently undergone a major codification wave, including a new Civil Transactions Law (721 articles), a new Companies Law (281 articles), a new Evidence Law, and a new Personal Status Law — no such resource has, to our knowledge, been publicly documented. Saudi legislation is published officially in the *Umm Al-Qura* gazette and on government portals as human-readable pages and PDF documents; it is not distributed as structured, article-level, provenance-annotated data.

This paper presents the Saudi Legal Corpus for AI, which closes that gap. The corpus structures the principal body of Saudi legislation into canonical article-level JSON with explicit provenance, wraps it in deterministic validation, and derives from it a unified retrieval index, a chunking layer, citation and supersession graphs, a glossary of statutory definitions, and a manually confirmed retrieval benchmark.

Our contributions are fourfold. First, the corpus itself: 290 legislative instruments and 15,689 article-level records, approximately 1.2 million Arabic tokens, spanning twelve legal domains from commercial and civil law through criminal procedure, labour, taxation, technology regulation, and the institutional statutes of major authorities (Section 5). Second, a methodology for auditable legal corpora: official-source grounding with per-record provenance labels, a four-tier source-verification taxonomy, a freshness manifest that documents known staleness risks instead of hiding them, and 376 read-only idempotent validators (Section 4). We argue that this methodology is a reusable template for legal corpora in any jurisdiction that lacks a machine-readable official gazette. Third, derived analytical layers unavailable elsewhere for Saudi law: a citation graph, a hand-classified supersession graph, and a statutory-definition glossary (Section 6). Fourth, a retrieval evaluation resource: 519 manually confirmed gold queries with two deterministic baselines reported per query category (Section 7).

Throughout, the corpus maintains a strict legal-status boundary. It is not an official government publication, it contains no official translation, and the official Arabic text governs; English and Chinese layers are non-authoritative references. We return to these boundaries in Section 10.

2 Related work

2.1 Legal corpora and benchmarks

Large legal text collections and benchmarks now exist for English and several European jurisdictions: the Pile of Law (Henderson et al., 2022), MultiLegalPile (Niklaus et al., 2024), LexGLUE (Chalkidis et al., 2022), LEXTREME (Niklaus et al., 2023), CUAD (Hendrycks et al., 2021), and LegalBench (Guha et al., 2023). These resources are dominated by English and European languages; none includes Saudi legislation, and Arabic is at most marginally represented.

Table 1 positions the Saudi Legal Corpus for AI against these resources. It is smaller by raw volume — deliberately so, since it is a structured statute corpus rather than a pretraining dump

Resource	Type	Languages	Unit	Provenance
Pile of Law	pretraining	English	document	no
MultiLegalPile	pretraining	24 (no Arabic)	document	per source
LexGLUE	benchmark	English	task example	no
LEXTREME	benchmark	24 (no Arabic)	task example	no
LegalBench	benchmark	English	task example	no
Saudi Legal Corpus for AI	corpus + eval	Arabic (gov.); EN/ZH ref.	article	per record

Table 1: Positioning against existing legal corpora and benchmarks. “gov.” denotes the governing layer; “ref.” denotes non-authoritative reference layers (English and Chinese currently cover one law). Per-record provenance here means 251 distinct source-status labels organized under a four-tier verification taxonomy.

— and it is the only one that is simultaneously article-level, Arabic-governing, and provenance-annotated at the level of the individual record.

2.2 Arabic legal NLP

Arabic NLP infrastructure has matured considerably (e.g., Antoun et al., 2020; Obeid et al., 2020), and domain-specific models such as AraLegal-BERT (AL-Qurishi et al., 2022) have been trained on proprietary Arabic legal text. Openly documented *data resources* for Arabic law — in particular structured, provenance-annotated statute corpora — nevertheless remain scarce. To our knowledge, no prior published resource provides article-level, machine-readable coverage of Saudi legislation with documented official-source provenance.

2.3 Legal document standards and citation networks

Standards such as Akoma Ntoso (Palmirani and Vitali, 2011) and the European Legislation Identifier define structural markup for legislation. Our canonical JSON is deliberately simpler — flat article-level records with rich retrieval metadata — but is convertible to such standards. A separate literature applies network science to legal citation structure (Fowler et al., 2007; Katz and Bommarito, 2014; Coupette et al., 2021); the citation and supersession graphs we release (Section 6) are, to our knowledge, the first machine-readable inputs enabling that line of work for Saudi, and more broadly Gulf, legislation.

3 Source material: the Saudi legislative landscape

Saudi legislation is enacted primarily by royal decree (*nizām*, conventionally rendered “law” or “system”) and elaborated by implementing regulations issued by ministers or authorities, alongside procedural rules, ministerial decisions, and official model forms. The authoritative publication channel is the official gazette *Umm Al-Qura*; consolidated texts are additionally published on official portals, principally the Bureau of Experts at the Council of Ministers (`laws.boe.gov.sa`) and the Ministry of Justice legal portal (`laws.moj.gov.sa`), as well as on the sites of issuing ministries and authorities.

Three unit terms recur in this paper and in the data at different granularities, and we fix them here. A **track** is the registry’s unit of management: one onboarded instrument-layer bundle

with its own sources, validators, and reports (291 tracks, of which 290 correspond to legislative instruments and one is a repository-level audit artifact). An **instrument** (`law_id`) is one enacted law, regulation, or rule set as identified in the unified index; 283 appear there. A **corpus** (`corpus_key`) is a subject-matter grouping that typically joins a law with its implementing regulation and annexes — the `labor` corpus, for example, spans eight components — of which there are 199.

Each track records its Arabic and English display names, issuing authority, official source URL or URLs, publication dates in both the Hijri and Gregorian calendars where available, language layers, record counts, and validator targets, in a central machine-readable registry (registry version 3.0).

4 Corpus design and methodology

4.1 Design principles

The corpus is built around four commitments.

The official Arabic text governs. Every record carries the official Arabic text verbatim (`text_ar`); no summarization, normalization, or editorial rewriting is applied to the governing text. All non-Arabic layers are explicitly flagged as non-authoritative.

Provenance is per-record and honest. Every record in the unified index carries a `text_status` provenance label describing exactly how its text was obtained and verified — for example, a cross-check between the Ministry of Justice portal application programming interface and the published official PDF. There are 251 distinct provenance labels, deliberately fine-grained rather than collapsed into a small ordinal scale, because collapsing provenance destroys auditability. Where a primary portal was found stale at build time, such as an amendment the portal had not yet incorporated, that fact is documented rather than silently patched: the freshness manifest flags 16 of 291 tracks as carrying a known source-staleness risk.

Validation is read-only, deterministic, and idempotent. The repository ships 376 validator targets, of 476 total build targets, that check schema conformance, record counts, hashes, layer boundaries, and cross-layer consistency without mutating any data. Derived-layer generators make no network calls and consult no wall-clock time, so every layer is exactly reproducible from the committed sources.

Derived layers are additive projections. The unified index, chunking layer, graphs, glossary, verification tiers, and freshness manifest are all read-only projections over the canonical track files. None of them alters legal text, and each records what it was generated from.

4.2 Source verification tiers

Because Saudi Arabia has no machine-readable official gazette, source quality varies by track and must be represented rather than assumed. Every track is assigned to one of four verification tiers (Table 2). The taxonomy allows downstream users to filter by evidential strength: a production RAG deployment may restrict itself to Tiers 1 and 2, while a coverage-oriented study may accept all four with the recorded caveats attached.

Tier	Tracks
1 — primary sources, multiple, in agreement	127
2 — primary source cross-verified against secondary	88
3 — secondary sources only, multiple, in agreement	39
4 — single source, or mixed confidence	37
Total	291

Table 2: Source-verification tier assignments across all tracks.

```
{
  "record_id": "aawan-regulation-llm-art-001",
  "corpus": "aawan",
  "law_id": "sa-aawan-regulation-1435",
  "law_component": "regulation",
  "law_title_ar": "اللائحة المنظمة لأعمال أعوان القضاء",
  "article_number": 1,
  "keywords_ar": "... يقصد، بأعوان، القضاء، ...",
  "search_queries_ar": "... المادة 1 اللائحة المنظمة لأعمال أعوان القضاء، ...",
  "text_ar": "... يقصد بأعوان القضاء: من يعين الدائرة في عملها المنصوص عليه نظاماً.",
  "text_status": "MOJ_PORTAL_API_CROSS_CHECKED_OFFICIAL_PDF",
  "source_layer": "aawan_regulation_legal_llm_001_035.json"
}
```

Figure 1: An example unified-index record (Article 1 of the Regulation of Judicial Assistants’ Work, abridged: the keyword and query lists are truncated). The article text translates as “Judicial assistants means: those who assist the circuit in its work as prescribed by law.” The `text_status` label records that the text was cross-checked between the Ministry of Justice portal interface and the published official PDF.

4.3 Record schema

The atomic unit is the article-level record (Figure 1). In the unified retrieval index each record carries a stable `record_id`; corpus and track identifiers (`corpus`, `law_id`, `law_component`); the official Arabic law title and the article number; generated retrieval metadata (`llm_title_ar`, `retrieval_title_ar`, `keywords_ar`, `search_queries_ar`); the verbatim governing text (`text_ar`); the provenance label (`text_status`); and the source layer file from which it was projected. Records are self-contained: a single retrieved record carries sufficient context to cite the law, the component, and the article.

4.4 Metadata generation and disclosure of automated assistance

Resource papers increasingly document not only what a dataset contains but how each field came to be (Bender and Friedman, 2018; Gebru et al., 2021). We therefore state the generation provenance of each field class explicitly. The governing Arabic text (`text_ar`) is ingested verbatim from the recorded official sources and is never generated, summarized, or normalized; validators enforce this by hash and count. The retrieval metadata is produced mechanically by each track’s deterministic generator script — keywords by term-frequency extraction from the article’s own text, search queries and retrieval titles by number and title templates — and not by a generative model, which

Corpus family	Tracks
Statutory laws	201
Statutory regulations	64
Implementing regulations	24
Procedural rules	1
Legislative instruments	290
Repository-level audit artifact	1
Total registry tracks	291

Table 3: Composition of the corpus by instrument type. The audit artifact is a quality-assurance record, not legislation, and carries no legal text.

Measure	Value
Article-level records	15,689
Distinct legislative instruments	283
Distinct subject-matter corpora	199
Arabic tokens (whitespace-delimited)	1,206,097
Characters	7,123,487
Distinct provenance labels	251

Table 4: The unified retrieval index at a glance.

is why the metadata-based searcher evaluated in Section 7 is deterministic end to end. The non-governing reference layers, namely the English layers and the internal Chinese layer, were produced with machine-assisted translation and passed a documented internal quality-assurance programme; they are explicitly non-official and non-binding. Repository tooling and documentation were developed with AI assistance; every data-bearing layer that this tooling produces is regenerable deterministically and checked by read-only validators.

5 Corpus contents and statistics

All figures reported in this section are computed by a read-only statistics script released with the corpus and are reproducible from the archived release cited in the data-availability statement.

5.1 Composition by instrument type

Table 3 gives the composition by instrument type. The registry counts 16,753 records in total: 15,858 primary Arabic governing records, 614 reference records in the English layers, and 281 internal reference records in the Chinese layer for the Companies Law. The unified retrieval index projects 15,689 of the primary governing records; the remaining 169 belong to a separate implementing-regulations intake track family that is registry-counted but served through its own layer files.

5.2 The unified retrieval index

Table 4 summarizes the unified index and Table 5 breaks its records down by component type. The long tail of component types — procedural manuals, model contract forms, recruitment rules,

Component	Records
Law articles	8,384
Regulation articles	4,655
Implementing-regulation articles	2,113
Procedural manuals	135
Model contract forms	102
Model work regulation	75
Recruitment services rules	72
Other rules, guides, mechanisms, annexes	153
Total	15,689

Table 5: Unified-index records by component type.

Legal domain	Instruments	Records
Courts, procedure and enforcement	42	3,377
Commercial and corporate	32	1,605
Financial and capital markets	30	1,272
Health, environment and safety	29	946
Civil, personal status and property	28	1,681
Energy, industry and infrastructure	28	1,766
Criminal justice, security and civil status	26	1,038
Technology, data, telecommunications and IP	20	633
Education, culture, media and civil society	20	963
Labour and social insurance	16	1,377
Tax and zakat	11	699
Constitutional and administrative	8	332
Total	290	15,689

Table 6: Coverage by legal domain under an author-assigned grouping (see text). Domain membership is assigned by published, inspectable rules and is not a property of the legislation itself.

accessibility arrangements — reflects a deliberate decision to ingest the official annexes that accompany major laws rather than the statute text alone, since those annexes carry operative content that practitioners consult directly.

5.3 Coverage by legal domain

The registry records no domain field, since domain membership is not a property of a Saudi legislative instrument. To convey coverage breadth to the reader we therefore applied an author-assigned grouping, implemented as explicit ordered keyword rules over track and corpus identifiers and released with the corpus so that it can be inspected and revised. Every track and every indexed record falls into exactly one of the twelve domains of Table 6; nothing is left unclassified, and the grouping should be read as an organizing device for this paper rather than as a legal taxonomy.

Coverage is broadest in adjudication and procedure, which is expected: the procedural instruments are long (the Sharia Procedure regulation alone contributes 637 articles) and each is accompanied by implementing rules. The comparatively small record counts in the technology and constitutional domains reflect the brevity of those instruments rather than gaps in coverage; the Personal Data Protection Law, for instance, is a complete track at 43 articles plus its regulations.

5.4 Multilingual layers

The Saudi Companies Law (Royal Decree M/132, 1443H; 281 articles) is the first track to receive full multilingual treatment: the governing Arabic layer, a full English layer of 281 records, an English reference layer of 281 records, and an internal Chinese reference layer that passed a documented four-phase remediation and quality-assurance programme closed by a repository-level audit. The corpus explicitly does not claim trilingual alignment or an official translation; the multilingual layers exist to support cross-lingual access research under the governing-text principle.

6 Derived analytical layers

6.1 Citation graph

A deterministic extractor scans all 15,689 records for statutory citations, yielding 3,607 references: 3,022 within a single instrument and 585 between distinct instruments. Each reference is emitted with the raw citation text, source and target identifiers, and a confidence label; 3,087 are high-confidence and 520 medium-confidence, with no ambiguous-scope cases. The 585 inter-instrument edges span 191 distinct source tracks and 316 distinct cited target names. The most frequently cited targets are the Labour Law with 34 inbound citations, the Companies Law with 30, the Bankruptcy Law with 19, and the Sharia Procedure Law with 14 — a distribution that already suggests which instruments function as structural hubs of the Saudi legal system. This is, to our knowledge, the first machine-readable citation network over Saudi legislation, and it directly enables network-analytic studies of centrality, clustering, and dangling references to superseded instruments.

6.2 Supersession graph

A separate layer records repeal and replacement relationships between instruments: 102 edges, each classified by hand against the corpus’s own documented text, namely registry notes and official-source metadata, and never inferred from the relative age of decrees or from topical similarity. Ambiguous cases, and title collisions that are not repeal relationships, are kept in dedicated sections rather than forced into the edge set. Combined with the citation graph, this layer supports temporal legal dynamics research, such as identifying live instruments that still cite repealed ones.

6.3 Statutory-definition glossary

Saudi laws conventionally open with a definitions article. A parser over these articles yields a corpus-wide glossary of 1,920 terms carrying 3,318 definitions, extracted from the 214 tracks whose definitions articles parse cleanly. Because the same term — “establishment” or “competent authority”, for instance — is defined independently by many laws, the glossary is the raw material for studying definitional fragmentation and terminological consistency across a national legal system.

6.4 Freshness manifest

A deterministic, network-free manifest surveys every track’s recorded source URLs, fetch dates, and known discrepancies, and flags 16 tracks as carrying a documented risk that the primary portal’s consolidated text was stale at build time. A separate live checker, explicitly outside the quality-assurance gate, can re-probe reachability and drift without compromising the determinism of the committed layers.

7 Retrieval index, chunking, and baseline evaluation

7.1 Chunking layer

For embedding pipelines, a chunking layer segments the unified index into 16,304 chunks sized in whitespace-delimited words and deliberately not tied to any particular embedding tokenizer. Of the source records, 15,344 fit within a single chunk and 345 long articles are split; each chunk carries its source record identifier, article number, chunk index, and exact character offsets into the source text, so that any vector hit remains traceable to its governing article. No embeddings are computed within the repository: the layer is pure, traceable segmentation.

7.2 Gold-query evaluation set

The retrieval evaluation pack contains 519 Arabic queries, each paired with a manually confirmed gold article identified by corpus, component, and article number. Golds were confirmed against the articles’ own text or official structure rather than reverse-engineered from search output, so the metrics measure the searcher honestly. The queries are categorized: 384 topical, 61 provision-lookup, 55 definitional, and 19 distributed across smaller categories covering governance, rights, procedure, and deliberately hard cases.

The queries and golds were authored by a single annotator, the corpus maintainer, by reading each article’s committed text first and writing the query from its own wording. There is consequently no inter-annotator agreement to report, and the set inherits an in-vocabulary bias that we quantify below. Provision-lookup queries such as “Article 5 of the Labour Law” share their surface form with the templated `search_queries_ar` metadata, so for that category the evaluation partly measures template matching rather than free-form retrieval. The per-category breakdown that follows keeps this visible instead of averaging it away.

7.3 Baseline results

We report two deterministic baselines. The *metadata searcher* is the corpus’s own lexical scorer over each record’s generated retrieval metadata, using weighted keyword, title, and query matching with no embeddings and no learned parameters. *BM25* is standard Okapi BM25 with $k_1 = 1.5$ and $b = 0.75$ over each record’s law title, article title, and verbatim text, with light Arabic orthographic normalization — diacritics and tatweel stripped, and alef, teh-marbuta, and alef-maqsurā variants folded — and whitespace tokenization, without stemming or a stopword list. Table 7 reports overall and per-category results; the 19 small-category queries are included in the overall row but not broken out.

Query set	Metadata searcher		BM25	
	Top-1	MRR@5	Top-1	MRR@5
All (519)	93.3	.956	90.4	.937
Topical (384)	93.2	.957	92.7	.956
Provision lookup (61)	98.4	.984	98.4	.984
Definitional (55)	90.9	.937	74.5	.809

Table 7: Top-1 accuracy (per cent) and mean reciprocal rank at rank 5 on the 519-query evaluation set, comparing the corpus’s metadata-based searcher against standard BM25 over titles and verbatim text.

Three observations follow. First, the near-identical provision-lookup scores confirm the template-matching effect disclosed above: both systems resolve “Article n of law X ” queries almost perfectly, so that category carries little information about retrieval quality. Second, the gap between the two systems concentrates precisely where retrieval is hardest. On definitional queries — where a query such as “definition of the sale contract” shares few surface tokens with the defining article — the engineered metadata is worth 16.4 points of top-1 accuracy over BM25. Third, six queries defeat the metadata searcher entirely, falling outside the top five; per-query results for both systems are published, which makes error analysis reproducible.

The remaining headroom, and more importantly robustness to paraphrase and dialectal variation that the gold queries do not exercise, is precisely the space in which neural Arabic legal retrievers should be evaluated. We release the queries, the golds, the per-query results, and both runner scripts to support that comparison.

8 Intended uses

The corpus is designed to serve five broad uses. As *grounding for retrieval-augmented Saudi-law assistants*, its per-record provenance and verification tiers allow deployments to filter by evidential strength rather than treating all text as equally trustworthy. As a resource for *Arabic legal NLP research*, it supports retrieval, summarization, question answering, terminology extraction, and cross-lingual legal access over a morphologically rich and highly formulaic register of Modern Standard Arabic. For *computational legal studies*, the released graphs enable citation-network analysis, studies of definitional consistency, work on the relationship between laws and their implementing regulations, and analysis of the temporal dynamics of the recent codification wave. As *ground truth for factuality benchmarking*, it provides the reference against which model hallucination on Saudi legal questions can be measured. Finally, the construction methodology is offered as a *template* for building auditable legal corpora in other jurisdictions lacking a machine-readable official gazette.

9 Limitations

The corpus is not exhaustive. Its 290 instruments cover the principal legislation but not every ministerial decision, circular, or municipal instrument; coverage is documented per track rather than claimed complete.

Consolidation risk is real and only partly mitigated. Official portals themselves sometimes

lag amendments; 16 tracks carry a documented staleness-risk flag and 37 tracks sit in the lowest verification tier. The corpus documents these risks, but documentation cannot eliminate them.

The extraction layers are heuristic. The citation extractor and the glossary parser are deterministic but pattern-based, and their published caveats — confidence labels and skipped tracks — bound but do not remove extraction error. The supersession graph is hand-classified and therefore conservative and possibly incomplete.

The evaluation set is single-annotator and lexically close to the corpus. Its 519 gold queries were authored by one annotator from the articles’ own wording, so no inter-annotator agreement can be reported, and the provision-lookup category overlaps the templated query metadata. The set measures in-vocabulary retrieval well and paraphrase robustness less well.

Multilingual depth is uneven. Only the Companies Law carries full English and internal Chinese layers; the corpus is Arabic-first by design.

Finally, the domain grouping in Section 5 is author-assigned. It is published as inspectable rules precisely because it embodies judgement calls that another researcher might make differently.

10 Ethical and legal considerations

The corpus contains enacted national legislation, which is a body of public legal fact rather than personal data. Three boundaries are nevertheless enforced repository-wide and restated here: the corpus is not an official government publication; its non-Arabic layers are not official translations; and nothing in it constitutes legal advice, the only binding text being the Arabic original as published in *Umm Al-Qura*.

The structuring apparatus — schemas, scripts, derived layers, metadata, and documentation — is released under the MIT licence. The underlying statutory texts remain public legal enactments of the Kingdom of Saudi Arabia; official texts of laws and regulations are excluded from copyright protection under Saudi copyright law, and the corpus asserts no ownership over them, reproducing them verbatim with per-record source attribution. Field-level generation provenance is documented in Section 4.4, in the spirit of data statements and datasheets (Bender and Friedman, 2018; Gebru et al., 2021).

A misuse consideration specific to legal corpora deserves emphasis: over-trust. Shipping provenance labels, verification tiers, and staleness flags inside the data, rather than in documentation that users never read, is a deliberate design choice intended to let downstream systems surface uncertainty instead of laundering it.

11 Conclusion and future work

We have presented the Saudi Legal Corpus for AI: 290 legislative instruments and 15,689 article-level records of Saudi legislation across twelve legal domains, with per-record provenance, tiered source verification, deterministic validation, derived citation, supersession, and glossary layers, and a manually confirmed 519-query retrieval benchmark with two deterministic baselines that already localize where retrieval is hard.

Future work proceeds along three axes. The first is neural baselines: Arabic and multilingual

embedding models evaluated on the released benchmark, together with paraphrase-robustness extensions of the query set and, ideally, a second annotator to establish agreement. The second is analytical studies over the released graphs — legislative network structure, definitional fragmentation, and the temporal dynamics of the Saudi codification wave. The third is coverage growth, onboarding further instruments through the same profile-driven factory, with amendment tracking remaining the hardest open problem for any legal corpus in a jurisdiction without a machine-readable gazette.

Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. The study involves no human participants, no animal subjects, and no personal data; it processes published national legislation only.

Data availability. The corpus, all derived layers, all validators, the evaluation pack, and the scripts that reproduce every figure reported in this paper are openly available at <https://github.com/Abdullah-m7/saudi-legal-corpus-ai> and archived on Zenodo under the MIT licence. The version described here is v1.0.2, **DOI: 10.5281/zenodo.22019183**; the concept DOI 10.5281/zenodo.22019182 always resolves to the latest version.

Code availability. All code is included in the archived release cited above.

Author contributions. A. Almohammed is the sole author and conducted all aspects of the work.

References

- Muhammad AL-Qurishi, Sarah AlQaseemi, and Riad Soussi. AraLegal-BERT: A pretrained language model for Arabic legal text. In *Proceedings of the Natural Legal Language Processing Workshop 2022*, Abu Dhabi, United Arab Emirates, 2022. Association for Computational Linguistics.
- Wissam Antoun, Fady Baly, and Hazem Hajj. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT)*, Marseille, France, 2020. European Language Resources Association.
- Emily M. Bender and Batya Friedman. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604, 2018.
- Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. LexGLUE: A benchmark dataset for legal language understanding

- in English. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, Dublin, Ireland, 2022. Association for Computational Linguistics.
- Corinna Coupette, Janis Beckedorf, Dirk Hartung, Michael Bommarito, and Daniel Martin Katz. Measuring law over time: A network analytical framework with an application to statutes and regulations in the United States and Germany. *Frontiers in Physics*, 9, 2021.
- James H. Fowler, Timothy R. Johnson, James F. Spriggs, Sangick Jeon, and Paul J. Wahlbeck. Network analysis and the law: Measuring the legal importance of precedents at the U.S. Supreme Court. *Political Analysis*, 15(3):324–346, 2007.
- Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.
- Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Ré, et al. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.
- Peter Henderson, Mark S. Krass, Lucia Zheng, Neel Guha, Christopher D. Manning, Dan Jurafsky, and Daniel E. Ho. Pile of Law: Learning responsible data filtering from the law and a 256GB open-source legal dataset. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.
- Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. CUAD: An expert-annotated NLP dataset for legal contract review. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021), Datasets and Benchmarks Track*, 2021.
- Daniel Martin Katz and Michael J. Bommarito. Measuring the complexity of the law: The United States Code. *Artificial Intelligence and Law*, 22(4):337–374, 2014.
- Joel Niklaus, Veton Matoshi, Pooja Rani, Andrea Galassi, Matthias Stürmer, and Ilias Chalkidis. LEXTREME: A multi-lingual and multi-task benchmark for the legal domain. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Singapore, 2023. Association for Computational Linguistics.
- Joel Niklaus, Veton Matoshi, Matthias Stürmer, Ilias Chalkidis, and Daniel E. Ho. MultiLegalPile: A 689GB multilingual legal corpus. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Bangkok, Thailand, 2024. Association for Computational Linguistics.
- Ossama Obeid, Nasser Zalmout, Salam Khalifa, Dima Taji, Mai Oudah, Bashar Alhafni, Go Inoue, Fadhil Eryani, Alexander Erdmann, and Nizar Habash. CAMEL Tools: An open source Python toolkit for Arabic natural language processing. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, Marseille, France, 2020. European Language Resources Association.
- Monica Palmirani and Fabio Vitali. Akoma-Ntoso for legal documents. In *Legislative XML for the Semantic Web*, pages 75–100. Springer, 2011.